BUSINESS REPORT

Health Care Capstone Project Notes 2

By

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1. Recap

Earlier in the stage of this project. Exploratory Data Analysis is conducted on the dataset and came to understanding of attributes relations with target variable which is Insurance Cost.
Picked Top 15 Features from the data by some Clustering techniques.

The Top 15 Features which has very strong importance when compared to Target Variable are these:

1. years_of_insurance_with_us
2. regular_checkup_lasy_year
3. visited_doctor_last_1_year
4. cholesterol_level
5. age
6. avg_glucose_level
7. bmi
8. smoking_status
9. weight
10. Alcohol
11. exercise
12. weight_change_in_last_one_year
13. fat_percentage
14. Gender_Male
15. covered_by_any_other_company_Y

These features are considered for the Next phase of the project which is Model Building.

2. Data Preparation

Splitting the Data into Train and Test which are in 70 and 30 proportions respectively.

```
Number of rows in train data = 17500
Number of rows in test data = 7500
```

3. Model Building

Total 8 Models are planned to build on this dataset.

1. 4 Parametric Models
    a. OLS Model
    b. LASSO Regression Model
    c. Ridge Regression Model
    d. Polynomial Regression Model
2. 4 Non-Parametric Models
    a. Random Forest Regression Model
    b. XG Boost Regression Model
    c. AdaBoost Regression Model
    d. Support Vector Machine (SVM) Regression Model

3.1 Parametric Models

3.1.1 OLS Model

```
                            OLS Regression Results
==============================================================================
Dep. Variable:         insurance_cost   R-squared:                       0.945
Model:                            OLS   Adj. R-squared:                  0.944
Method:                 Least Squares   F-statistic:                 1.985e+04
Date:                Sun, 03 Aug 2025   Prob (F-statistic):               0.00
Time:                        17:50:44   Log-Likelihood:            -1.6705e+05
No. Observations:               17500   AIC:                         3.341e+05
Df Residuals:                   17484   BIC:                         3.343e+05
Df Model:                          15
Covariance Type:            nonrobust
==================================================================================================
                                     coef    std err          t      P>|t|      [0.025      0.975]
--------------------------------------------------------------------------------------------------
const                          -8.013e+04    277.964   -288.259      0.000   -8.07e+04   -7.96e+04
years_of_insurance_with_us       -13.0167     10.177     -1.279      0.201     -32.965       6.931
cholesterol_level                 25.0007     20.445      1.223      0.221     -15.074      65.076
age                                2.8229      1.589      1.776      0.076      -0.292       5.938
avg_glucose_level                 56.8511     30.982      1.835      0.067      -3.877     117.580
bmi                               26.8032     41.435      0.647      0.518     -54.413     108.019
weight                          1490.4753      2.992    498.106      0.000    1484.610    1496.340
weight_change_in_last_one_year   175.2827     16.348     10.722      0.000     143.239     207.326
fat_percentage                   -29.4794     29.989     -0.983      0.326     -88.262      29.303
regular_checkup_lasy_year       -452.4039     21.587    -20.957      0.000    -494.717    -410.091
visited_doctor_last_1_year       -33.9938     22.600     -1.504      0.133     -78.291      10.304
smoking_status                    -6.7683     21.281     -0.318      0.750     -48.481      34.945
Alcohol                           20.9754     40.718      0.515      0.606     -58.836     100.787
exercise                          -9.3513     31.877     -0.293      0.769     -71.834      53.131
Gender_Male                       31.1523     59.094      0.527      0.598     -84.677     146.982
covered_by_any_other_company_Y  1209.4524     57.756     20.941      0.000    1096.245    1322.660
==============================================================================
Omnibus:                      549.476   Durbin-Watson:                   1.980
Prob(Omnibus):                  0.000   Jarque-Bera (JB):              650.857
Skew:                           0.394   Prob(JB):                    4.66e-142
Kurtosis:                       3.520   Cond. No.                         931.
==============================================================================

Notes:
[1] Standard Errors assume that the covariance matrix of the errors is correctly specified.
```

Figure 1: Model Summary

3.1.2 Model Performance on Training data

Training Performance

	RMSE	MAE	R-squared	Adj. R-squared	MAPE	Model
0	3383.976844	2726.866303	0.94454	0.944489	15.364108	OLS Model

Figure 2: Model Summary

Model Evaluation checks

1. TEST FOR MULTICOLLINEARITY

    a. If VIF is between 1 and 5, then there is low multicollinearity.
    b. If VIF is between 5 and 10, we say there is moderate multicollinearity.
    c. If VIF is exceeding 10, it shows signs of high multicollinearity.

	feature	VIF
0	const	117.968023
1	years_of_insurance_with_us	1.074730
2	cholesterol_level	1.012511
3	age	1.000604
4	avg_glucose_level	1.001233
5	bmi	1.181960
6	weight	1.195902
7	weight_change_in_last_one_year	1.171035
8	fat_percentage	1.010097
9	regular_checkup_lasy_year	1.024778
10	visited_doctor_last_1_year	1.004790
11	smoking_status	1.094231
12	Alcohol	1.004201
13	exercise	1.001658
14	Gender_Male	1.200788
15	covered_by_any_other_company_Y	1.081473

Figure 3: VIF Summary

- There are no multiple columns with very high VIF values, indicating no presence of multicollinearity

- We will systematically drop numerical columns with VIF > 5
- We will ignore the VIF values for dummy variables and the constant (intercept)

2. **Linearity of variables**
3. **Independence of error terms**
4. **Normality of error terms**
5. **No Heteroscedasticity**

	Actual Values	Fitted Values	Residuals
4289	28382.0	29604.903540	-1222.903540
19621	23446.0	20769.153934	2676.846066
14965	46892.0	44276.790919	2615.209081
12321	40722.0	45462.870559	-4740.870559
6269	32084.0	35857.236859	-3773.236859

Figure 4: Predicted values

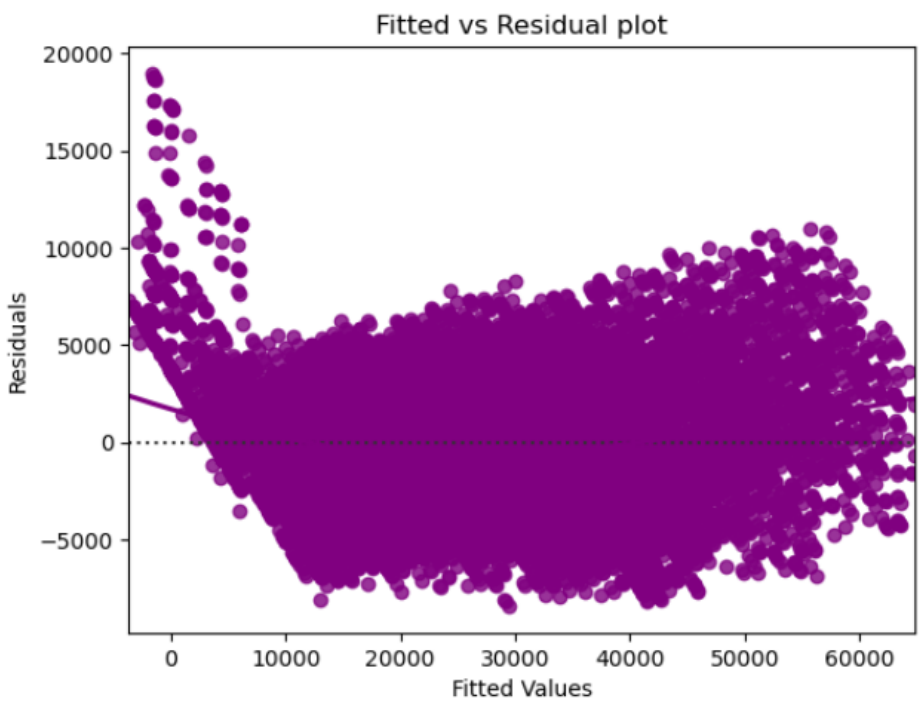


Figure 5: Residual Chart

- The scatter plot shows the distribution of residuals (errors) vs fitted values (predicted values).

- If there exist any pattern in this plot, we consider it as signs of non-linearity in the data and a pattern means that the model doesn't capture non-linear effects.
- **We see no pattern in the plot above. Hence, the assumptions of linearity and independence are satisfied.**

6. TEST FOR NORMALITY

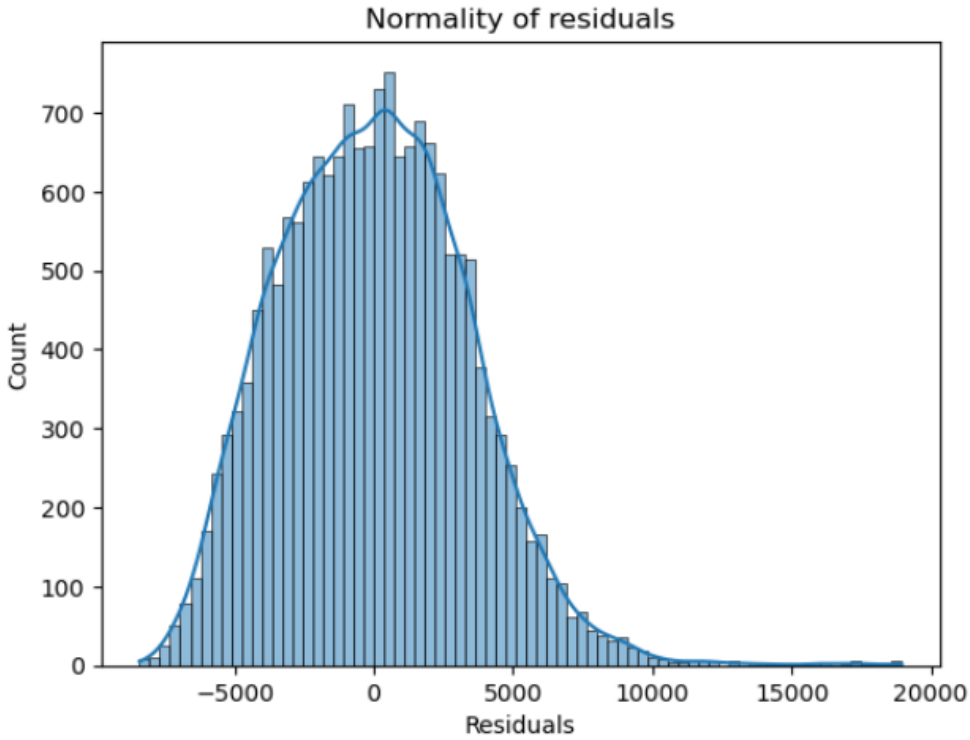


Figure 6: Normality of Residuals

- The histogram of residuals does have a bell shape.
- Let us check the Q-Q plot.

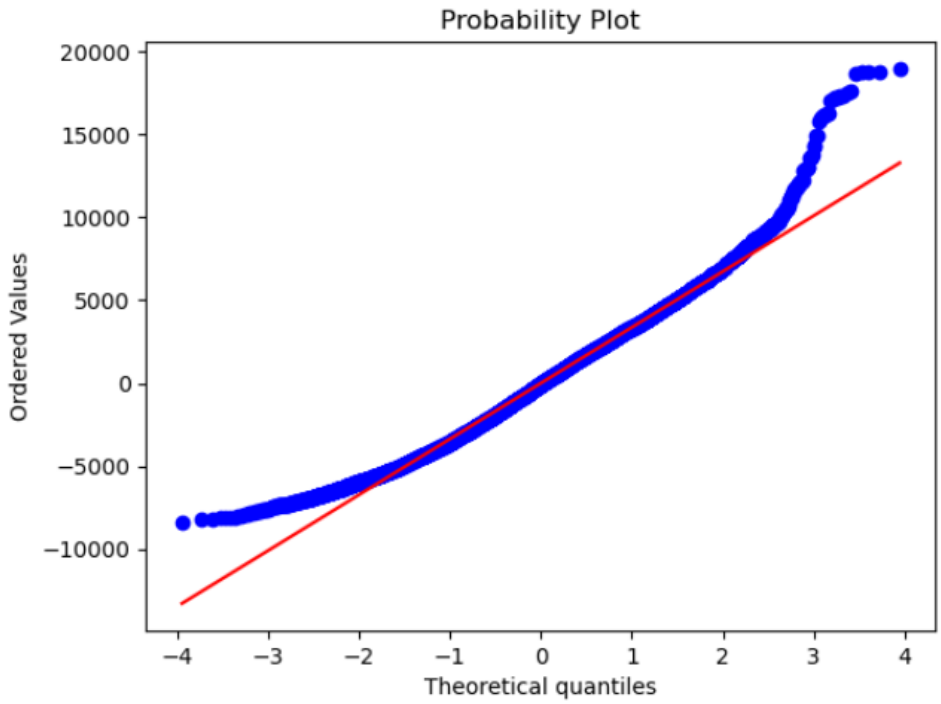


Figure 7: Q-Q Plot

7. Shapiro Test

```
ShapiroResult(statistic=0.987649019540111, pvalue=5.266053488552379e-36)
```

- Since p-value < 0.05, the residuals are not normal as per the Shapiro-Wilk test.
- Strictly speaking, the residuals are not normal.
- However, as an approximation, we can accept this distribution as close to being normal.
- **So, the assumption is satisfied.**

8. F Statistic Test

```
[('F statistic', 1.0007029442440367), ('p-value', 0.4869032216102235)]
```

Figure 8: F-Statistic

Since p-value > 0.05, we can say that the residuals are homoscedastic. So, this assumption is satisfied.

3.1.3 Model Performance on Test data

```
Test Performance
```

	RMSE	MAE	R-squared	Adj. R-squared	MAPE	Model
0	3340.182673	2711.95957	0.94479	0.944672	15.115261	OLS Model

Figure 9: Model Summary

- The model can explain ~94% of the variation in the data

3.1.4 LASSO Regression Model

LASSO Coefficients and Intercepts

```
Lasso Coefficients: [   0.          -12.26374286   23.49492635    2.81254998   53.72218469
   24.29975337 1490.45354718  174.38205263  -26.46086157 -450.99606236
  -32.33464418   -6.57755496   15.78393108   -6.23876337   23.62149231
 1198.54543901]
Intercept: -80115.19393698919
```

Figure 10: Model Coefficients

3.1.5 Model Performance on Training data

```
Training Performance
```

	RMSE	MAE	R-squared	Adj. R-squared	MAPE	Model
0	3383.989207	2726.856992	0.944539	0.944488	15.362929	LASSO Regression Model

Figure 11: Model Summary

3.1.6 Model Performance on Test data

```
Test Performance
```

	RMSE	MAE	R-squared	Adj. R-squared	MAPE	Model
0	3339.870964	2711.725453	0.9448	0.944682	15.11247	LASSO Regression Model

Figure 12: Model Summary

3.1.7 Ridge Regression Model

Ridge Coefficients and Intercepts

```
Ridge Coefficients: [   0.          -13.00049281   25.00009673    2.82287846   56.84577733
   26.80010695 1490.47528828  175.27379223  -29.4757048  -452.38385525
  -33.99088784   -6.76972836   20.9725242    -9.35072696   31.14397109
 1209.09990176]
Intercept: -80125.79032924659
```

Figure 13: Model Coefficients

3.1.8 Model Performance on Training data

Training Performance

	RMSE	MAE	R-squared	Adj. R-squared	MAPE	Model
0	3383.976847	2726.865727	0.94454	0.944489	15.364054	Ridge Regression Model

Figure 14: Model Summary

3.1.9 Model Performance on Test data

Test Performance

	RMSE	MAE	R-squared	Adj. R-squared	MAPE	Model
0	3340.181495	2711.959904	0.94479	0.944672	15.115224	Ridge Regression Model

Figure 15: Model Summary

3.1.10 Polynomial Regression Model

Polynomial Coefficients and Intercepts with Degree 2

```
Polynomial Coefficients: [-2.32620418e-12  3.05801649e+13 -1.49093017e+13  2.68806813e+13
  2.79347620e+12 -8.04821461e+11 -7.23866105e+10 -5.50793322e+11
 -9.51081988e+11  1.09306005e+11  1.63064589e+12 -5.15544525e+11
 -9.11311039e+11  4.10417096e+11 -3.43810583e+10 -1.55761462e+11
 -1.15743111e+11  4.03096959e+11  1.49093017e+13 -2.68806813e+13
 -2.79347620e+12  8.04821461e+11  7.23866105e+10  5.50793323e+11
  9.51081988e+11 -1.09306005e+11 -1.63064589e+12  5.15544525e+11
  9.11311038e+11 -4.10417096e+11  3.43810588e+10 -5.73185327e+09
  8.15123762e+10 -3.28271484e+00  7.18719482e+00 -1.38476562e+00
 -1.21018066e+01  3.53593750e+01 -6.12890625e+00  2.00186768e+01
 -9.99511719e-01 -8.35086060e+00  1.12084961e+00  1.85058594e+00
  2.94323120e+01  7.28900146e+00 -2.24451752e+01  2.42598114e+01
  3.11630707e+01 -1.45770264e+00  9.47769928e+00  1.33621063e+01
  6.02783203e-01  1.06763611e+01  2.66733246e+01  9.40368652e+00
 -3.17717056e+01  4.15069313e+01 -2.93419552e+01 -3.08427734e+01
  8.73083115e+00 -1.93857994e+01  1.10351562e-01  7.19604492e-01
 -3.16162109e+00  2.53906250e-01  2.92968750e-03  3.72070312e-01
  1.22070312e-02  1.11816406e+00 -1.55419922e+00  3.18237305e-01
  8.93554688e-01 -3.25756836e+00  1.22314453e-01 -2.58117065e+01
  6.38427734e-01 -2.49218750e+00 -1.55403137e+01 -4.87280884e+01
  1.19592590e+01 -9.64092064e+00  2.09856300e+01 -1.07066861e+02
  8.18051910e+01  3.93762474e+01  1.01526005e+02  5.08978882e+01
  2.55078125e+00  7.19604492e+00 -9.19572449e+00 -1.82461548e+00
 -3.76477661e+01 -9.71563721e+00 -6.86494751e+01 -2.24437714e+01
 -2.31682892e+01 -1.23584734e+02  7.03515625e+00 -5.39062500e+00
  2.27148438e+00 -1.48691406e+01  1.50683594e+00  2.69628906e+00
 -5.94482422e-01 -6.99267578e+00 -3.97753906e+00  8.74555664e+01
  9.59450073e+01 -6.37927246e+01 -5.21328430e+01 -2.62469482e+00
 -2.46665955e-01 -4.52156830e+00 -2.71109009e+00 -3.10994453e+01
 -1.57767212e+02 -1.53644409e+01 -4.95735168e+00  2.09503174e-01
  6.29699707e-01 -7.11988525e+01  1.28529510e+01 -4.47697067e+01
 -5.86489677e+01  9.05541382e+01 -3.68533936e+01  3.62471619e+01
 -3.15697327e+01  1.24545135e+01  1.76093903e+01  2.21622314e+01
  5.27673340e+00  1.81825867e+01 -7.32446289e+00 -5.81988525e+01
  2.96691895e-01  1.28589935e+01 -4.63595581e+00  1.79979248e+01
  4.13490982e+01  1.64613190e+01 -2.71899223e+01 -3.36017838e+01
  5.06747246e+01  1.24331081e+02 -2.99902935e+01 -5.62976227e+01
  2.47227287e+01  1.14948166e+02  1.61493316e+11  1.54255295e+01
  3.42307299e+10]
Intercept: -30983261909551.254
```

Figure 16: Model Coefficients

3.1.11 Model Performance on Training data

Training Performance

	RMSE	MAE	R-squared	Adj. R-squared	MAPE	Model
0	3211.688318	2608.852823	0.950043	0.949997	13.984243	Polynomial Regression Model

Figure 17: Model Summary

3.1.12 Model Performance on Test data

Test Performance

	RMSE	MAE	R-squared	Adj. R-squared	MAPE	Model
0	3200.639262	2612.016973	0.949306	0.949198	13.906082	Polynomial Regression Model

Figure 18: Model Summary

3.1.13 Model Performance Comparison on Training data between Parametric Models

Training performance comparison:

	RMSE	MAE	R-squared	Adj. R-squared	MAPE	Model
0	3383.976844	2726.866303	0.944540	0.944489	15.364108	OLS Model
1	3383.989207	2726.856992	0.944539	0.944488	15.362929	LASSO Regression Model
2	3340.181495	2711.959904	0.944790	0.944672	15.115224	Ridge Regression Model
3	3211.688318	2608.852823	0.950043	0.949997	13.984243	Polynomial Regression Model

Figure 19: Model Summary

3.1.14 Model Performance Comparison on Test data between Parametric Models

Testing performance comparison:

	RMSE	MAE	R-squared	Adj. R-squared	MAPE	Model
0	3340.182673	2711.959570	0.944790	0.944672	15.115261	OLS Model
1	3339.870964	2711.725453	0.944800	0.944682	15.112470	LASSO Regression Model
2	3383.976847	2726.865727	0.944540	0.944489	15.364054	Ridge Regression Model
3	3200.639262	2612.016973	0.949306	0.949198	13.906082	Polynomial Regression Model

Figure 20: Model Summary

Polynomial Regression Model with Degree 2 has Lesser MAPE and Higher Adjusted R Squared compared to other Parametric Models.

3.2 Non-Parametric Models

Data is split into 3 sets, Train, Validation, Test.

```
Number of rows in train data = 15000
Number of rows in validation data = 5000
Number of rows in test data = 5000
```

Figure 21: Data Split

3.2.1 Model Performance Comparison on Training and Validation data

```
Training Performance:

Random forest:
          RMSE         MAE  R-squared  Adj. R-squared      MAPE
0  1168.241102  926.231351   0.993361        0.993354  4.543973

XGB:
          RMSE          MAE  R-squared  Adj. R-squared      MAPE
0  2307.866801  1821.500487   0.974091        0.974063  9.071529

Adaboost:
          RMSE          MAE  R-squared  Adj. R-squared       MAPE
0  3250.350122  2690.963374   0.948608        0.948553  15.848071

SVR:
           RMSE          MAE  R-squared  Adj. R-squared      MAPE
0  14058.819429  11595.92706    0.03854        0.037514  78.59003


Validation Performance:

Random forest:
          RMSE          MAE  R-squared  Adj. R-squared       MAPE
0  3124.925289  2502.941944   0.952283        0.952129  12.339513

XGB:
          RMSE          MAE  R-squared  Adj. R-squared       MAPE
0  3146.491594  2531.833429   0.951622        0.951466  12.594991

Adaboost:
          RMSE          MAE  R-squared  Adj. R-squared       MAPE
0  3280.273472  2705.700624    0.94742        0.947251  15.927634

SVR:
          RMSE           MAE  R-squared  Adj. R-squared       MAPE
0  14025.58842  11577.663506   0.038741        0.035655  78.701528
```

Figure 22: Model Summary

3.2.2 Model Performance Differences on Training and Validation data

```
Training and Validation Performance Difference:

Random forest:
  RMSE  - Train: 1168.24, Validation: 3124.93, Diff: -1956.68
  MAE   - Train: 926.23, Validation: 2502.94, Diff: -1576.71
  R²    - Train: 0.99, Validation: 0.95, Diff: 0.04
  Adj R²- Train: 0.99, Validation: 0.95, Diff: 0.04
  MAPE  - Train: 4.54%, Validation: 12.34%, Diff: -7.80%

XGB:
  RMSE  - Train: 2307.87, Validation: 3146.49, Diff: -838.62
  MAE   - Train: 1821.50, Validation: 2531.83, Diff: -710.33
  R²    - Train: 0.97, Validation: 0.95, Diff: 0.02
  Adj R²- Train: 0.97, Validation: 0.95, Diff: 0.02
  MAPE  - Train: 9.07%, Validation: 12.59%, Diff: -3.52%

Adaboost:
  RMSE  - Train: 3250.35, Validation: 3280.27, Diff: -29.92
  MAE   - Train: 2690.96, Validation: 2705.70, Diff: -14.74
  R²    - Train: 0.95, Validation: 0.95, Diff: 0.00
  Adj R²- Train: 0.95, Validation: 0.95, Diff: 0.00
  MAPE  - Train: 15.85%, Validation: 15.93%, Diff: -0.08%

SVR:
  RMSE  - Train: 14058.82, Validation: 14025.59, Diff: 33.23
  MAE   - Train: 11595.93, Validation: 11577.66, Diff: 18.26
  R²    - Train: 0.04, Validation: 0.04, Diff: -0.00
  Adj R²- Train: 0.04, Validation: 0.04, Diff: 0.00
  MAPE  - Train: 78.59%, Validation: 78.70%, Diff: -0.11%
```

Figure 23:Model Summary

3.2.3 All Model Performance Differences on Training data

Model	RMSE	MAE	R-squared	Adj. R-squared	MAPE
Random forest	1168.241102	926.231351	0.993361	0.993354	4.543973
XGB	2307.866801	1821.500487	0.974091	0.974063	9.071529
Adaboost	3250.350122	2690.963374	0.948608	0.948553	15.848071
SVR	14058.819429	11595.927060	0.038540	0.037514	78.590030

Figure 24: Model Summary

Models like Random Forest, XGB, AdaBoost are good with MAPE but the models might be overfitted. This raises the need for Hyperparameter Tuning all the models.

3.2.4 Tuned Random Forest Regression Model Performance on Train and Validation data

```
 Final Scores for Best Estimator:
Training Set:
          RMSE         MAE  R-squared  Adj. R-squared      MAPE  \
0  1227.742216  983.113595   0.992668         0.99266  5.137801

                Model
0  Tuned RandomForest

Validation Set:
          RMSE          MAE  R-squared  Adj. R-squared       MAPE
0  3247.031591  2608.949572   0.948481        0.948315  13.687654
CPU times: total: 10.6 s
Wall time: 1min 32s
```

Figure 25: Model Summary

3.2.5 Tuned XG Boost Regression Model Performance on Train and Validation data

```
Best Parameters Found:
{'subsample': 0.8, 'n_estimators': 175, 'max_depth': 3, 'learning_rate': 0.05}

 Performance Metrics – Best XGBoost Estimator
Training:
          RMSE          MAE  R-squared  Adj. R-squared       MAPE  \
0  2974.037953  2399.201729   0.956974        0.956929  11.939725

           Model
0  Tuned XGBoost

Validation:
          RMSE          MAE  R-squared  Adj. R-squared      MAPE
0  3004.052406  2418.236994   0.955903        0.955761  11.90582
```

Figure 26: Model Summary

3.2.6 Tuned Adaboost Regressor Model Performance on Train and Validation data

```
Best Parameters:
{'n_estimators': 175, 'loss': 'exponential', 'learning_rate': 0.05}

 Performance Metrics – Best AdaBoost Model
Training:
          RMSE          MAE  R-squared  Adj. R-squared       MAPE  \
0  3194.328316  2594.425703   0.950364        0.950311  13.977834

            Model
0  Tuned Adaboost

Validation:
          RMSE          MAE  R-squared  Adj. R-squared       MAPE
0  3228.724341  2611.744489    0.94906        0.948896  14.041136
```

Figure 27: Model Summary

3.2.7 Tuned SVR Regressor Model Performance on Train and Validation data

```
Best Parameters:
{'svr__kernel': 'linear', 'svr__gamma': 'scale', 'svr__epsilon': 0.1, 'svr__C': 10}

 Performance Metrics – Best SVR Model
Training:
          RMSE          MAE  R-squared  Adj. R-squared       MAPE      Model
0  3399.810658  2723.705345   0.943773        0.943713  15.113603  Tuned SVR

Validation:
          RMSE          MAE  R-squared  Adj. R-squared       MAPE
0  3414.964466  2723.795565   0.943014        0.942831  15.062678
```

Figure 28: Model Summary

3.2.8 Model Performance Comparison on Training data between Non-Parametric Models

Model	RMSE	MAE	R-squared	Adj. R-squared	MAPE
Tuned RandomForest	1227.742216	983.113595	0.992668	0.992660	5.137801
Tuned XGBoost	2974.037953	2399.201729	0.956974	0.956929	11.939725
Tuned Adaboost	3194.328316	2594.425703	0.950364	0.950311	13.977834
Tuned SVR	3399.810658	2723.705345	0.943773	0.943713	15.113603

Figure 29: Model Summary

3.2.9 All Model Performance Comparison on Training data between Non-Parametric Models

Model	RMSE	MAE	R-squared	Adj. R-squared	MAPE
Random forest	1168.241102	926.231351	0.993361	0.993354	4.543973
XGB	2307.866801	1821.500487	0.974091	0.974063	9.071529
Adaboost	3250.350122	2690.963374	0.948608	0.948553	15.848071
SVR	14058.819429	11595.927060	0.038540	0.037514	78.590030
Tuned RandomForest	1227.742216	983.113595	0.992668	0.992660	5.137801
Tuned XGBoost	2974.037953	2399.201729	0.956974	0.956929	11.939725
Tuned Adaboost	3194.328316	2594.425703	0.950364	0.950311	13.977834
Tuned SVR	3399.810658	2723.705345	0.943773	0.943713	15.113603

Figure 30: Model Summary

3.2.10 Model Performance Comparison on Test data between Non-Parametric Models

Model	RMSE	MAE	R-squared	Adj. R-squared	MAPE
Tuned RandomForest	1227.742216	983.113595	0.992668	0.992660	5.137801
Tuned XGBoost	2974.037953	2399.201729	0.956974	0.956929	11.939725
Tuned Adaboost	3194.328316	2594.425703	0.950364	0.950311	13.977834
Tuned SVR	3399.810658	2723.705345	0.943773	0.943713	15.113603

Figure 31: Model Summary

3.2.11 All Model Performance Comparison on Test data between All Models

Model	RMSE	MAE	R-squared	Adj. R-squared	MAPE
OLS Model	3383.976844	2726.866303	0.944540	0.944489	15.364108
LASSO Regression Model	3383.989207	2726.856992	0.944539	0.944488	15.362929
Ridge Regression Model	3340.181495	2711.959904	0.944790	0.944672	15.115224
Polynomial Regression Model	3211.688318	2608.852823	0.950043	0.949997	13.984243
Tuned RandomForest	1227.742216	983.113595	0.992668	0.992660	5.137801
Tuned XGBoost	2974.037953	2399.201729	0.956974	0.956929	11.939725
Tuned Adaboost	3194.328316	2594.425703	0.950364	0.950311	13.977834
Tuned SVR	3399.810658	2723.705345	0.943773	0.943713	15.113603

Figure 32: Model Summary

3.2.12 All Model Performance Comparison on Train data between All Models Sorted

Model	RMSE	MAE	R-squared	Adj. R-squared	MAPE
Tuned RandomForest	1227.742216	983.113595	0.992668	0.992660	5.137801
Tuned XGBoost	2974.037953	2399.201729	0.956974	0.956929	11.939725
Tuned Adaboost	3194.328316	2594.425703	0.950364	0.950311	13.977834
Polynomial Regression Model	3211.688318	2608.852823	0.950043	0.949997	13.984243
Tuned SVR	3399.810658	2723.705345	0.943773	0.943713	15.113603
Ridge Regression Model	3340.181495	2711.959904	0.944790	0.944672	15.115224
LASSO Regression Model	3383.989207	2726.856992	0.944539	0.944488	15.362929
OLS Model	3383.976844	2726.866303	0.944540	0.944489	15.364108

Figure 33: Model Summary

3.2.13 All Model Performance Comparison on Test data between All Models Sorted

Model	RMSE	MAE	R-squared	Adj. R-squared	MAPE
Tuned XGBoost	3108.266238	2510.930901	0.952741	0.952589	12.257640
Tuned Random Forest	3333.187473	2694.652893	0.945654	0.945480	13.758945
Polynomial Regression Model	3200.639262	2612.016973	0.949306	0.949198	13.906082
Tuned Adaboost	3302.679856	2678.215204	0.946644	0.946473	14.240940
LASSO Regression Model	3339.870964	2711.725453	0.944800	0.944682	15.112470
OLS Model	3340.182673	2711.959570	0.944790	0.944672	15.115261
Ridge Regression Model	3383.976847	2726.865727	0.944540	0.944489	15.364054
Tuned SVR	3486.833863	2800.861362	0.940528	0.940337	15.443473

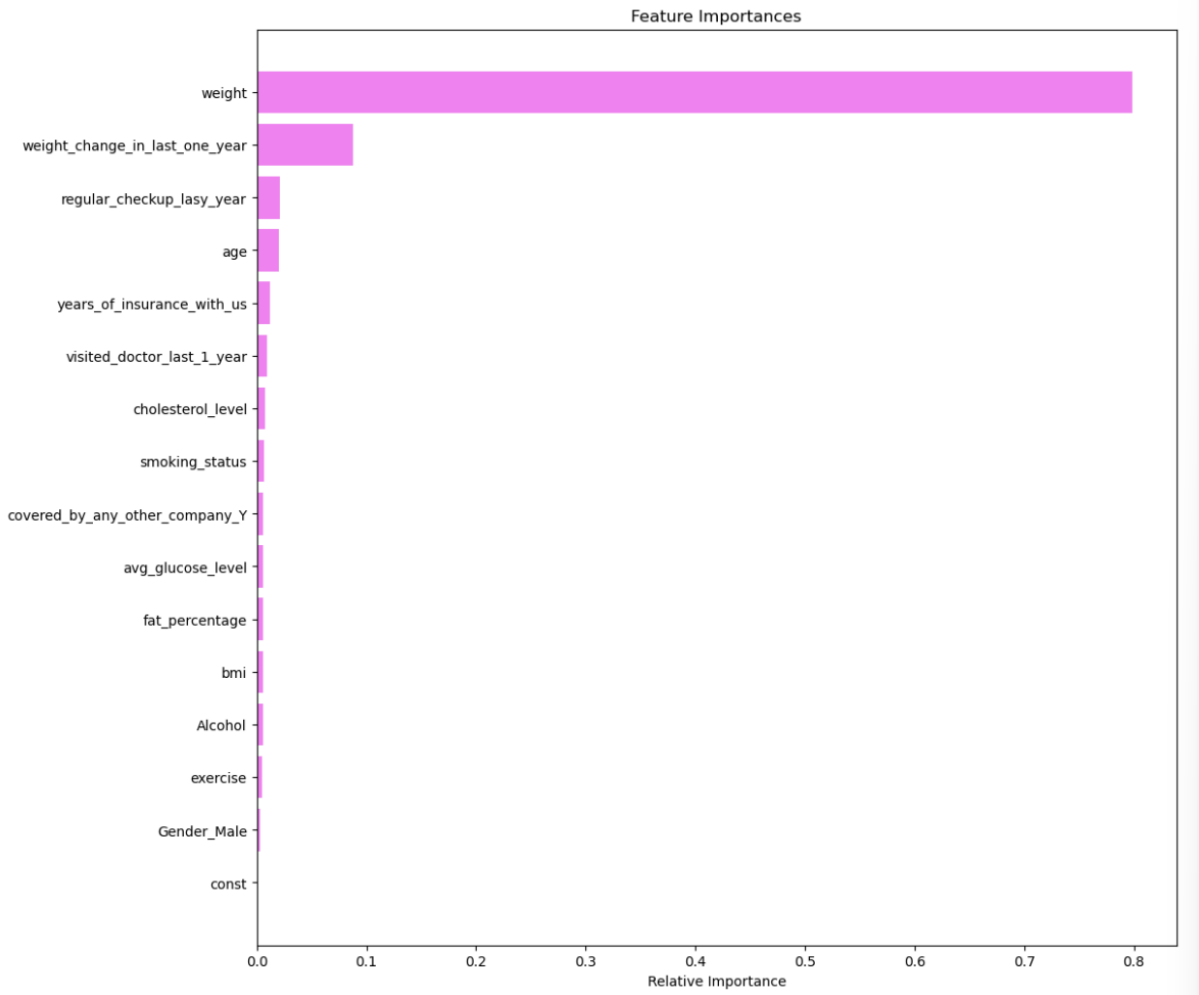
Figure 34: Model Summary

All models have given MAPE less than 15% on test data compared with Train data

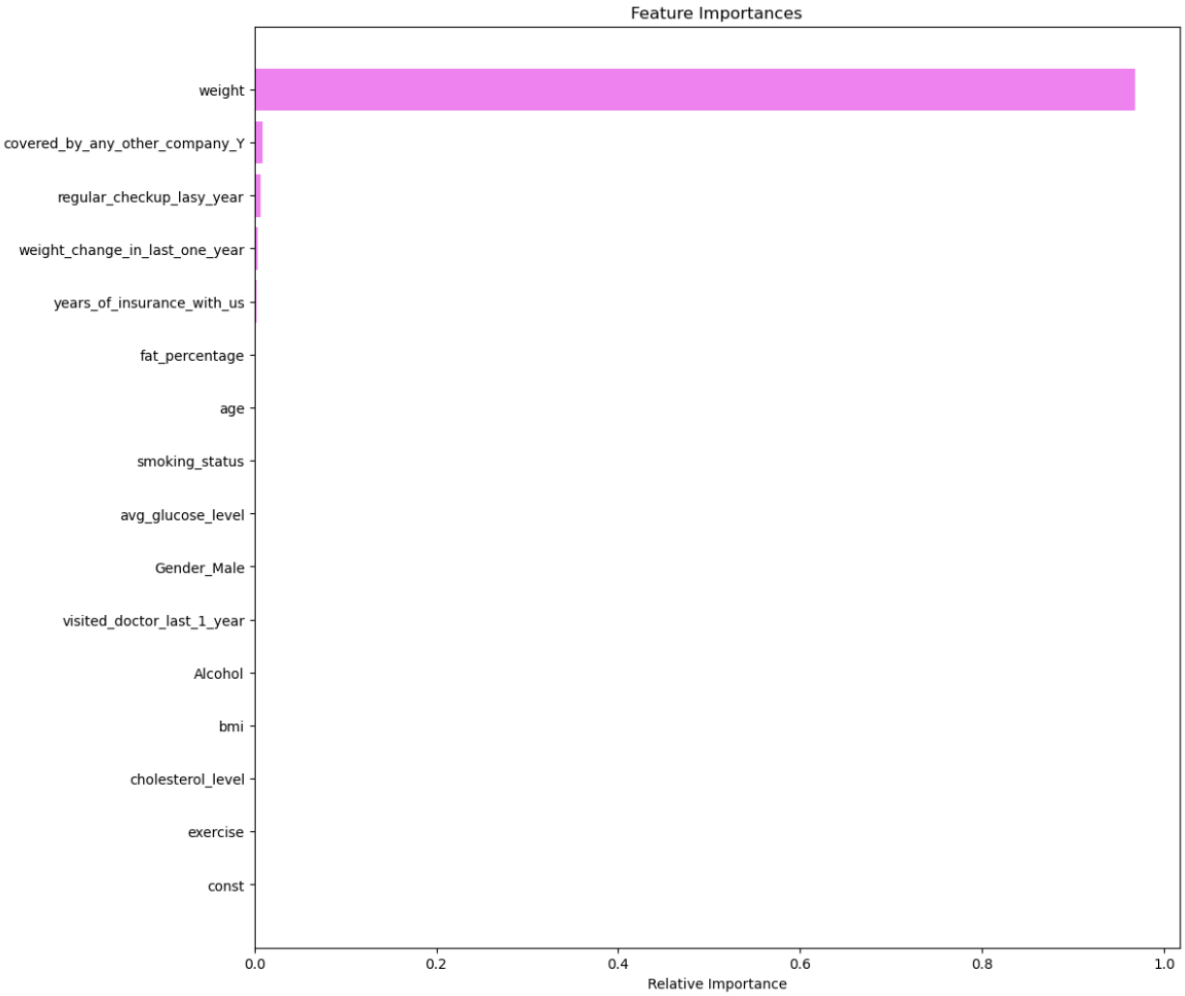
This performance is in line with what we achieved with this model on the train and validation sets

So, picking top 2 models to check feature importances

3.3 Feature Importance of Tuned Random Forest Model



3.4 Feature Importance of Tuned XG Boost Model



3.5 Interpretation of the Most Optimum model

Tuned XG Boost:

- The Tuned XG Boost model stands out as the most accurate and business-aligned choice in lineup. With a MAPE of just 12.26%, it offers exceptionally tight forecasting performance.
- Its R-squared value of 0.9527 confirms that it captures over 95% of the variance in the target variable, signifying strong feature.
- The model also posted the lowest RMSE and MAE, smoother performance across diverse customer profiles.
- In terms of feature behaviour, XG Boost places overwhelming emphasis on “weight”. Other features like coverage by another insurance company and regular health checkups register minor but intuitive signals, highlighting behavioural and contextual flags.

Tuned Random Forest:

- The Tuned XG Boost model stands out as the slightly lesser accurate than Tuned XG Boost model. With a MAPE of just 13.76%, it offers exceptionally tight forecasting performance.
- Its R-squared value of 0.9457 confirms that it captures over 94% of the variance in the target variable, this model still delivers highly dependable performance — just a notch below XG Boost.
- The slightly higher RMSE and MAE suggest broader prediction intervals but remain well within acceptable thresholds.
- In terms of feature behaviour, Random Forest truly shines in its clarity of feature importances. Not only does it reaffirm **weight** as the primary driver ~80% importance, but it also surfaces **weight change in the last year** as a meaningful secondary signal ~10%. Other features like **checkups**, **age**, **tenure**, and **lifestyle indicators** contribute minimally